Type MS · Sum and product of roots

Answer 1 of 21 · Question 11 part b.i · 2 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part b.i in this mark scheme extract

11 (a)(i)	$\left(\alpha-\beta=2\sqrt{6}\Rightarrow(\alpha-\beta)^2=24\right)\Rightarrow(\alpha+\beta)^2-4\alpha\beta=24$ oe **(1)**	M1
	$\alpha^2+\beta^2=30\Rightarrow(\alpha+\beta)^2-2\alpha\beta=30$ oe **(2)**	M1
	(2) – (1) $6=2\alpha\beta\Rightarrow\alpha\beta=3$ cso	dM1A1* [4]
ALT1	$\left((\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta\Rightarrow\right)2\alpha\beta=\alpha^2+\beta^2-(\alpha-\beta)^2$ oe	M1
	$30-\left(2\sqrt{6}\right)^2=6$	M1
	$2\alpha\beta=6\Rightarrow\alpha\beta=3$	dM1A1 [4]
(ii)	$30=(\alpha+\beta)^2-2\times3\Rightarrow(\alpha+\beta)^2=36\Rightarrow\alpha+\beta=6$ $[\alpha>\beta>0]$ cso	M1A1* [2]
ALT2 (i)	$\alpha-\beta=2\sqrt{6}\rightarrow\alpha=2\sqrt{6}+\beta$	
	$\alpha^2+\beta^2=30\rightarrow\left(2\sqrt{6}+\beta\right)^2+\beta^2=30$	M1
	$2\beta^2+4\sqrt{6}\beta-6=0$ oe eg $\beta^2+2\sqrt{6}\beta-3=0$	M1
	$(\beta=)\dfrac{-2\sqrt{6}\pm\sqrt{\left(2\sqrt{6}\right)^2-4(1)(-3)}}{2}$ oe $\rightarrow\beta\left(=3-\sqrt{6}\right)$	dM1
	$\alpha=3+\sqrt{6}$	
	$\alpha\beta=\left(3+\sqrt{6}\right)\left(3-\sqrt{6}\right)=9+6\sqrt{3}-6\sqrt{3}-6=3$	A1* [4]
(ii)	$\alpha+\beta=3+\sqrt{6}+3-\sqrt{6}=6$	M1A1 [2]
(b)(i)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2(\alpha\beta)^2=30^2-2\times3^2=882$	M1A1 [2]
(b) ii)	$\alpha^4-\beta^4=\left(\alpha^2+\beta^2\right)\left(\alpha^2-\beta^2\right)=\left(\alpha^2+\beta^2\right)(\alpha-\beta)(\alpha+\beta)$	M1
	$\alpha^4-\beta^4=30\times6\times2\sqrt{6}=360\sqrt{6}$	A1 [2]
(c)	$\left(\alpha^4+\beta^4\right)+\left(\alpha^4-\beta^4\right)=2\alpha^4=882+360\sqrt{6}\Rightarrow\alpha^4=441+180\sqrt{6}$	M1A1 [2]
ALT1	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\alpha=3+\sqrt{6}$	
	$\alpha^4=\left(3+\sqrt{6}\right)^4$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
ALT2	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\beta=3-\sqrt{6}$	
	$\alpha^4=\beta^4+360\sqrt{6}=\left(3-\sqrt{6}\right)^4+360\sqrt{6}$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
		Total 12 marks

Type MS · Sum and product of roots

Answer 1 of 21 · Question 11 part b.i · 2 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part b.i in this mark scheme extract

Part	Mark	Notes
(a) (i)	M1	For forming an equation of the form $(\alpha+\beta)^2 \pm p\alpha\beta = 24$ oe or $(\alpha+\beta)^2 \pm t\alpha\beta = 30$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correctly solving their simultaneous equations in $\alpha+\beta$ and $\alpha\beta$ to find a value for $\alpha\beta$, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
ALT1	M1	For forming an equation of the form $q\alpha\beta = \alpha^2+\beta^2-(\alpha-\beta)^2$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correct substitution into their equation, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
(a) (ii)	M1	Correctly uses either of their equations (must be of the required form) to substitute the given value of $\alpha\beta$ and obtains a value for $\alpha+\beta$
	A1*cso	For the correct value of $\alpha+\beta$
ALT2 (i)	M1	For an attempt to eliminate α or β and arrive at an unsimplified quadratic equation in one variable. Allow one error
	M1	For the correct quadratic equation
	dM1	For a **fully correct method** to solve their quadratic equation in α or β, dependent on the 1st method mark
	A1*cso	For correctly finding α and β and showing the minimum steps shown to find $\alpha\beta$
	M1	For finding $\alpha+\beta$ with their values
	A1*cso	For finding $\alpha+\beta$ with the correct values, minimum steps as shown.
(b) (i)	M1	For the correct algebra $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2(\alpha\beta)^2$, if they use more complex algebra, it must be correct and fully ready for substitution of given values. Do not allow $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2\alpha^2\beta^2$ unless recovered
	A1	For substituting the given values to find the correct value for $\alpha^4+\beta^4$
(b) (ii)	M1	For the correct algebra to write $\alpha^4-\beta^4$ in terms of $(\alpha^2+\beta^2), (\alpha-\beta)$ and $(\alpha+\beta)$
	A1	For the correct value of $\alpha^4-\beta^4$
(c)	M1	For adding together $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate β^4 and reach α^4 either as an expression or implied by adding their values together and dividing by 2. If students subtract $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate α^4 and reach β^4, they must then reach α^4 as an expression or implied by subtracting their values, using one of the expressions from part b and dividing by 2.
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT1	M1	For $\alpha^4 = (3+\sqrt{6})^4$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT2	M1	For $\alpha^4 = \beta^4+360\sqrt{6} = (3-\sqrt{6})^4+360\sqrt{6}$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$

Type MS · Sum and product of roots

Answer 1 of 21 · Question 11 part b.i · 2 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part b.i in this mark scheme extract

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Type MS · Sum and product of roots

Answer 2 of 21 · Question 10 part a · 3 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
10(a)	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ $\frac{115}{8}=\left(-\frac{5}{2}\right)^3-3\alpha\beta\left(-\frac{5}{2}\right)$ $\Rightarrow \alpha\beta=\frac{-\frac{115}{8}+\left(-\frac{5}{2}\right)^3}{3\times\left(-\frac{5}{2}\right)}$ $\Rightarrow \alpha\beta=4$	M1 dM1 A1 cso [3]
(b)	**Sum:** $\frac{\alpha^2+1}{\beta}+\frac{\beta^2+1}{\alpha}=\frac{\alpha^3+\beta^3+\alpha+\beta}{\alpha\beta}$ $\Rightarrow \frac{\alpha^2+1}{\beta}+\frac{\beta^2+1}{\alpha}=\frac{\frac{115}{8}+\left(-\frac{5}{2}\right)}{4}=\frac{95}{32}$ **Product:** $\frac{\alpha^2+1}{\beta}\times\frac{\beta^2+1}{\alpha}=\frac{\alpha^2\beta^2+\alpha^2+\beta^2+1}{\alpha\beta}$ $\Rightarrow \frac{\alpha^2+1}{\beta}\times\frac{\beta^2+1}{\alpha}=\frac{\alpha^2\beta^2+\left[(\alpha+\beta)^2-2\alpha\beta\right]+1}{\alpha\beta}$ $\Rightarrow \frac{\alpha^2+1}{\beta}\times\frac{\beta^2+1}{\alpha}=\frac{4^2+\left(-\frac{5}{2}\right)^2-2\times4+1}{4}=\frac{61}{16}$ **Equation:** $x^2-\left(\frac{95}{32}\right)x+\left(\frac{61}{16}\right)=0\Rightarrow 32x^2-95x+122=0$ oe	M1A1 M1 M1 A1 M1A1 [7]
		Total 10 marks

Type MS · Sum and product of roots

Answer 2 of 21 · Question 10 part a · 3 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	Part (a) is a 'Show that' question. You must see sufficient work for the award of **both** M marks for the award of the A mark	
	M1	For the correct algebra **and** substitution of the given values into a correct expression for $\alpha^3+\beta^3$ There is more than one acceptable form of this expansion. They must be able to substitute the given values of $\alpha+\beta$ and $\alpha^3+\beta^3$ with $\alpha\beta$ as the value to find in any version they use. For example: $\alpha^3+\beta^3=(\alpha+\beta)\left[(\alpha+\beta)^2-3\alpha\beta\right]$
	dM1	For an attempt to solve the linear equation to find a value for $\alpha\beta$ Allow one processing error for this mark. Note, this is a dependent M mark.
	A1 cso	For $\alpha\beta=4$ This is a show question, you must check their algebra carefully.
(b)	M1	For the **correct** algebra **and** substitution of the given values to find the sum.
	A1	For the correct sum $=\frac{95}{32}$
	M1	For the correct algebra **and** substitution to find $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=\left[-\frac{7}{4}\right]$
	M1	For the **correct** algebra **and** substitution of the given values to find the product
	A1	For the correct product $=\frac{61}{16}$
	M1	For forming an equation using **their** sum and product correctly. $x^2-(\text{their sum})x+(\text{their product})=(0)$ Allow missing = 0 for this mark
	A1	For a correct equation with integer coefficients. For example: $64x^2-190x+244=0$

Type MS · Sum and product of roots

Answer 2 of 21 · Question 10 part a · 3 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum and product of roots

Answer 3 of 21 · Question 5 part a · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

5(a)	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=$ $\beta^3\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3-3\alpha^2\beta-3\alpha\beta^2=\alpha^3+\beta^3$	M1 A1cso [2]
ALT (a)	$(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ $=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$ $=(\alpha+\beta)\left(\alpha^2+\beta^2-\alpha\beta\right)$ $=\alpha^3+\alpha\beta^2-\alpha^2\beta+\beta\alpha^2+\beta^3-\beta^2\alpha$ $=\alpha^3+\beta^3$	 M1 A1 cso [2]
(b)	$\alpha\beta=-\frac{7}{2}$ and $\alpha+\beta=3$ Product: $\frac{\alpha^2}{\beta}\times\frac{\beta^2}{\alpha}=\alpha\beta=-\frac{7}{2}$ Sum: $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}=\left\{\frac{\alpha^3+\beta^3}{\alpha\beta}\right\}=\frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}=\frac{3^3-3\times-\frac{7}{2}\times3}{-\frac{7}{2}}=-\frac{117}{7}$ Equation: $x^2+\frac{117}{7}x-\frac{7}{2}=0\Rightarrow14x^2+234x-49=0$	B1 B1ft M1A1 M1A1 [6]
		Total 8 marks

Type MS · Sum and product of roots

Answer 3 of 21 · Question 5 part a · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correctly expanded $(\alpha+\beta)^3$ $(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$
	A1cso	For obtaining the given result without any errors
ALT(a)	M1	Factorises $(\alpha+\beta)$ first then attempts to expand all brackets
	A1cso	For obtaining the given result without any errors
(b)	B1	For the correct sum **and** product of the given equation Allow calling α,β as x_1,x_2 or other letters
	B1ft	For the product $\frac{\alpha^2}{\beta}\times\frac{\beta^2}{\alpha}=\alpha\beta=\text{'}-\frac{7}{2}\text{'}$, follows through their value of $\alpha\beta$
	M1	For the sum $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}=\frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}=\frac{(\text{'3'})^3-(3)\times\left(\text{'}-\frac{7}{2}\text{'}\right)\times(\text{'3'})}{\left(\text{'}-\frac{7}{2}\text{'}\right)}=\left(-\frac{117}{7}\right)$ Correct algebra and **correct** substitution of values of their $\alpha\beta$ and $\alpha+\beta$
	A1	For the correct sum $-\frac{117}{7}$
	M1	For correctly forming an equation with their sum and product $x^2-\text{sum}\ x+\text{product}=0$ $x^2\text{'}+\frac{117}{7}\text{'}x\text{'}-\frac{7}{2}\text{'}=0$ **Condone the absence of = 0** for this mark.
	A1	For the correct equation $14x^2+234x-49=0$ oe Must be integers coefficients

Type MS · Sum and product of roots

Answer 3 of 21 · Question 5 part a · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 4 of 21 · Question 4 part a · 4 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

4 (a)	$\alpha+\beta=-2 \qquad \alpha\beta=\frac{3}{2}$	B1
	$(\alpha+\beta)^2-2\alpha\beta=\alpha^2+\beta^2$	M1
	$(-2)^2-2\left(\frac{3}{2}\right)=4-3=1$ *	M1 A1cso (4)
ALT		
	$2\alpha^2+4\alpha+3=0$ and $2\beta^2+4\beta+3=0$	{B1}
	$2\ \alpha^2+\beta^2\ +4\ \alpha+\beta\ +6=0$	{M1}
	$\alpha^2+\beta^2\ =-2\ -2\ -3=1$*	{M1} {A1cso} (4)
(b)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)(\alpha^2+\beta^2)-2\alpha^2\beta^2$	M1
	$1\times1-2\left(\frac{3}{2}\right)^2=-\frac{7}{2}$	M1 A1 (3)
ALT		
	$\alpha^2+\beta^2\ ^2=1=\alpha^4+2\alpha^2\beta^2+\beta^4=\alpha^4+\beta^4+2\times\frac{9}{4}$	{M1}
	$\alpha^4+\beta^4=1-\frac{9}{2}=-\frac{7}{2}$	{M1} {A1} (3)
(c)	Product of the roots: $\alpha^4\beta^4=\left(\frac{3}{2}\right)^4=\frac{81}{16}$	M1
	(Sum of the roots: $\alpha^4+\beta^4=-\frac{7}{2}$)	
	$16x^2+56x+81=0$	M1 A1ft (3)
		Total 10 marks

Type MS · Sum and product of roots

Answer 4 of 21 · Question 4 part a · 4 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $\alpha+\beta=-2$ and $\alpha\beta=\frac{3}{2}$
	M1	For $(\alpha+\beta)^2-2\alpha\beta=\alpha^2+\beta^2$ This must be correct
	M1	For substituting their sum and product into their expansion for $\alpha^2+\beta^2$ $(-2)^2-2\left(\frac{3}{2}\right)$
	A1cso	For obtaining the given expression $\alpha^2+\beta^2=1$
	ALT	
	B1	For $2\alpha^2+4\alpha+3=0$ and $2\beta^2+4\beta+3=0$
	M1	For $2\ \alpha^2+\beta^2\ +4\ \alpha+\beta\ +6=0$
	M1	For $-2\ -2\ -3$
	A1cso	For obtaining the given expression $\alpha^2+\beta^2=1$
(b)	M1	For $\alpha^4+\beta^4=(\alpha^2+\beta^2)(\alpha^2+\beta^2)-2\alpha^2\beta^2$ This must be correct
	M1	For substituting $\alpha^2+\beta^2=1$ and their product from (a) into their expansion for $\alpha^4+\beta^4$ $\alpha^4+\beta^4=1\times1-2\left(\frac{3}{2}\right)^2$
	A1	For the correct value $\alpha^4+\beta^4=-\frac{7}{2}$
	ALT	
	M1	For $1=\alpha^4+2\alpha^2\beta^2+\beta^4$
	M1	For $1=\alpha^4+\beta^4+2\times\frac{9}{4}$
	A1	For $\alpha^4+\beta^4=-\frac{7}{2}$
(d)	M1	For product of the roots: $\alpha^4\beta^4\left(=\left(\frac{3}{2}\right)^4=\frac{81}{16}\right)$
	M1	For use of $x^2-(\text{their sum})\,x+(\text{their product})=[0]$
	A1	For $16x^2+56x+81=0$ Must include = 0

Type MS · Sum and product of roots

Answer 4 of 21 · Question 4 part a · 4 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 5 of 21 · Question 4 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

4 (a)	$\alpha+\beta=-2 \quad \alpha\beta=\frac{3}{2}$	B1
	$(\alpha+\beta)^2-2\alpha\beta=\alpha^2+\beta^2$	M1
	$(-2)^2-2\left(\frac{3}{2}\right)=4-3=1$ *	M1 A1cso (4)
ALT		
	$2\alpha^2+4\alpha+3=0$ and $2\beta^2+4\beta+3=0$	{B1}
	$2\ \alpha^2+\beta^2\ +4\ \alpha+\beta\ +6=0$	{M1}
	$\alpha^2+\beta^2\ =-2\ -2\ -3=1$*	{M1} {A1cso} (4)
(b)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)(\alpha^2+\beta^2)-2\alpha^2\beta^2$	M1
	$1\times1-2\left(\frac{3}{2}\right)^2=-\frac{7}{2}$	M1 A1 (3)
ALT		
	$\alpha^2+\beta^2\ ^2=1=\alpha^4+2\alpha^2\beta^2+\beta^4=\alpha^4+\beta^4+2\times\frac{9}{4}$	{M1}
	$\alpha^4+\beta^4=1-\frac{9}{2}=-\frac{7}{2}$	{M1} {A1} (3)
(c)	Product of the roots: $\alpha^4\beta^4=\left(\frac{3}{2}\right)^4=\frac{81}{16}$	M1
	(Sum of the roots: $\alpha^4+\beta^4=-\frac{7}{2}$)	
	$16x^2+56x+81=0$	M1 A1ft (3)
		Total 10 marks

Type MS · Sum and product of roots

Answer 5 of 21 · Question 4 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $\alpha+\beta=-2$ and $\alpha\beta=\frac{3}{2}$
	M1	For $(\alpha+\beta)^2-2\alpha\beta=\alpha^2+\beta^2$ This must be correct
	M1	For substituting their sum and product into their expansion for $\alpha^2+\beta^2$ $(-2)^2-2\left(\frac{3}{2}\right)$
	A1cso	For obtaining the given expression $\alpha^2+\beta^2=1$
	ALT	
	B1	For $2\alpha^2+4\alpha+3=0$ and $2\beta^2+4\beta+3=0$
	M1	For $2\ \alpha^2+\beta^2\ +4\ \alpha+\beta\ +6=0$
	M1	For $-2\ -2\ -3$
	A1cso	For obtaining the given expression $\alpha^2+\beta^2=1$
(b)	M1	For $\alpha^4+\beta^4=(\alpha^2+\beta^2)(\alpha^2+\beta^2)-2\alpha^2\beta^2$ This must be correct
	M1	For substituting $\alpha^2+\beta^2=1$ and their product from (a) into their expansion for $\alpha^4+\beta^4$ $\alpha^4+\beta^4=1\times1-2\left(\frac{3}{2}\right)^2$
	A1	For the correct value $\alpha^4+\beta^4=-\frac{7}{2}$
	ALT	
	M1	For $1=\alpha^4+2\alpha^2\beta^2+\beta^4$
	M1	For $1=\alpha^4+\beta^4+2\times\frac{9}{4}$
	A1	For $\alpha^4+\beta^4=-\frac{7}{2}$
(d)	M1	For product of the roots: $\alpha^4\beta^4\left(=\left(\frac{3}{2}\right)^4=\frac{81}{16}\right)$
	M1	For use of $x^2-(\text{their sum})\,x+(\text{their product})=[0]$
	A1	For $16x^2+56x+81=0$ Must include = 0

Type MS · Sum and product of roots

Answer 5 of 21 · Question 4 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 6 of 21 · Question 8 part a · 3 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

8(a)	$\alpha+\beta=\frac{k}{3}$ and $\alpha\beta=-\frac{1}{3}$	B1
	$(\alpha+\beta)^2=\alpha^2+2\alpha\beta+\beta^2\Rightarrow\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$	M1
	$\alpha^2+\beta^2=\left(\frac{k}{3}\right)^3-2\left(-\frac{1}{3}\right)=\frac{k^2}{9}+\frac{2}{3}=\frac{k^2+6}{9}*$	A1 cso [3]
(b)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2$	M1
	$\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2$	M1
	$\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2\Rightarrow k^4+12k^2-448=0$	M1
	$k^4+12k^2-448=(k^2-16)(k^2+28)=0\Rightarrow k^2=16\Rightarrow k=4$	M1A1 [5]
(c)	Sum: $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\alpha^4+\beta^4+2\alpha\beta}{\alpha\beta}$	M1
	$\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\frac{466}{81}+2\left(-\frac{1}{3}\right)}{-\frac{1}{3}}=-\frac{412}{27}$	A1
	Product: $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\alpha^4+\beta^4+(\alpha\beta)^3+\alpha\beta}{\alpha\beta}$	M1
	$\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\frac{466}{81}+\left(-\frac{1}{3}\right)^3+\left(-\frac{1}{3}\right)}{\left(-\frac{1}{3}\right)}=-\frac{436}{27}$	A1
	3TQ: $x^2+\frac{412}{27}x-\frac{436}{27}=0\Rightarrow 27x^2+412x-436=0$	M1A1 [6]
		Total 14 marks

Question	Scheme	Marks
8(a)	$\alpha+\beta=\frac{k}{3}$ and $\alpha\beta=-\frac{1}{3}$	B1
	For the correct algebra on $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$	M1
	For substituting *their* sum and *their* product to give a simplified expression for $\alpha^2+\beta^2=\left(\frac{k}{3}\right)^3-2\left(-\frac{1}{3}\right)=\frac{k^2}{9}+\frac{2}{3}=\frac{k^2+6}{9}*$	A1 cso [3]

Type MS · Sum and product of roots

Answer 6 of 21 · Question 8 part a · 3 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

(b)	For the correct algebra on $\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2\alpha^2\beta^2$ Or $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2$	M1
	For substituting the values for $\alpha^2+\beta^2$ (in terms of k) and $\alpha\beta$ into their $\alpha^4+\beta^4$ and equating to $\frac{466}{81}$ $\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2$ Or for substituting the values for $\alpha^2+\beta^2$ (in terms of k), $\alpha+\beta$ and $\alpha\beta$ into their $\alpha^4+\beta^4$ and equating to $\frac{466}{81}$	M1
	For forming a simplified quadratic. $\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2\Rightarrow k^4+12k^2-448=0$ Or for $\left(\frac{k^2+6}{9}\right)^2=\frac{448}{81}$ Their quadratic must come from an attempt at $\alpha^4+\beta^4$	M1
	For solving their quadratic. $k^4+12k^2-448=\left(k^2-16\right)\left(k^2+28\right)=0\Rightarrow k^2=16\Rightarrow k=4$ Their quadratic must come from an attempt at $\alpha^4+\beta^4$	M1
	For the correct value of $k=4$ If they give $k=\pm4$ this is A0	A1 [5]
(c)	For the correct algebra on the sum. $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\alpha^4+\beta^4+2\alpha\beta}{\alpha\beta}$	M1
	For the correct value of the sum. $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\frac{466}{81}+2\left(-\frac{1}{3}\right)}{-\frac{1}{3}}=-\frac{412}{27}$	A1
	For the correct algebra on the product. $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\alpha^4+\beta^4+(\alpha\beta)^3+\alpha\beta}{\alpha\beta}$	M1
	For the correct value of the product. $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\frac{466}{81}+\left(-\frac{1}{3}\right)^3+\left(-\frac{1}{3}\right)}{\left(-\frac{1}{3}\right)}=-\frac{436}{27}$	A1
	For the 3TQ from their values of the sum and product (used correctly) [= 0 not required for this mark].	M1

Type MS · Sum and product of roots

Answer 6 of 21 · Question 8 part a · 3 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

	$x^2+\frac{412}{27}x-\frac{436}{27}=(0)$	
	For the correct 3TQ. Must have $= 0$. $27x^2+412x-436=0$ Must have integer coefficients. Accept a multiple of this.	A1 [6]
		Total 14 marks

Question	Scheme	Marks

Type MS · Sum and product of roots

Answer 7 of 21 · Question 10 part a.i · 0 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

10(a)	$\alpha+\beta=-3 \quad \alpha\beta=-5$	B1
(i)	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \ =19$	M1,A1
(ii)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \ =19^2-50=311$	M1,A1(5)
	OR: $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2, \ =19^2-50=311$	
(b)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=19+10$ OR $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=9-(-20)$	M1
	$\alpha-\beta=\sqrt{29}$ *	A1 cso (2)
(c)	$\alpha^4-\beta^4=(\alpha^2-\beta^2)(\alpha^2+\beta^2)=(\alpha-\beta)(\alpha+\beta)(\alpha^2+\beta^2)$	M1A1A1 (3)
(d)	$\alpha^4-\beta^4=\sqrt{29}\times(-3)\times19=-57\sqrt{29} \ \left(-\sqrt{94221}\right)$	M1A1 (2)
(e)	$2\beta^4=\alpha^4+\beta^4-(\alpha^4-\beta^4)$	M1
	$\beta^4=\frac{1}{2}(311+57\sqrt{29}), =\frac{311}{2}+\frac{57}{2}\sqrt{29}$	A1,A1 (3)
	$p=\frac{311}{2} \quad q=\frac{57}{2}$	[15]
ALT	$\beta^4=\left(\frac{-3-\sqrt{29}}{2}\right)^4$ **and** use a correct binomial expansion M1A1 Correct final answer A1	

(a)B1	Correct sum **and** product of roots, seen explicitly or used (in (a)). Must be clear that sum is negative
(i)M1	Correct algebra, ready for substitution of sum and product
A1	Correct answer, condone use of $\alpha+\beta=3$.
(ii)M1	Correct algebra, ready for substitution
A1	Correct answer, condone use of $\alpha+\beta=3$.
(b)M1	Correct algebra and substitution of their values
A1cso	Correct answer from correct working. Must have seen sum = -3 here if not shown in (a)
(c)	
M1	Factorise to 2 quadratic brackets or 2 linear and one quadratic bracket
A1	Obtain 2 linear and 1 quadratic brackets with 2 of the 3 brackets correct
A1	Third correct bracket Accept $(\alpha^2+\beta^2)$ or $((\alpha+\beta)^2-2\alpha\beta)$
(d)M1	Substitute their values for each of the 3 brackets obtained in (c)
A1	Correct answer as shown or equivalent **exact** value
(e)	
M1	Correct expression for $2\beta^4$ or β^4
A1ft	Substitute their numbers to obtain a numerical expression for β^4 The expression must be exact but need not be simplified **NB** A correct numerical expression for their values implies M1
A1	Correct answer in the required form. p and q need not be shown explicitly.

Type MS · Sum and product of roots

Answer 7 of 21 · Question 10 part a.i · 0 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Sum and product of roots

Answer 8 of 21 · Question 10 part a.ii · 5 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number		
10(a)	$\alpha+\beta=-3 \quad \alpha\beta=-5$	B1
(i)	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \ =19$	M1,A1
(ii)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \ =19^2-50=311$	M1,A1(5)
	OR: $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2, \ =19^2-50=311$	
(b)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=19+10$ OR $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=9-(-20)$	M1
	$\alpha-\beta=\sqrt{29}$ *	A1 cso (2)
(c)	$\alpha^4-\beta^4=(\alpha^2-\beta^2)(\alpha^2+\beta^2)=(\alpha-\beta)(\alpha+\beta)(\alpha^2+\beta^2)$	M1A1A1 (3)
(d)	$\alpha^4-\beta^4=\sqrt{29}\times(-3)\times19=-57\sqrt{29} \ \left(-\sqrt{94221}\right)$	M1A1 (2)
(e)	$2\beta^4=\alpha^4+\beta^4-(\alpha^4-\beta^4)$	M1
	$\beta^4=\frac{1}{2}(311+57\sqrt{29}), =\frac{311}{2}+\frac{57}{2}\sqrt{29}$	A1,A1 (3)
	$p=\frac{311}{2} \quad q=\frac{57}{2}$	[15]
ALT	$\beta^4=\left(\frac{-3-\sqrt{29}}{2}\right)^4$ **and** use a correct binomial expansion M1A1 Correct final answer A1	

(a)B1	Correct sum **and** product of roots, seen explicitly or used (in (a)). Must be clear that sum is negative
(i)M1	Correct algebra, ready for substitution of sum and product
A1	Correct answer, condone use of $\alpha+\beta=3$.
(ii)M1	Correct algebra, ready for substitution
A1	Correct answer, condone use of $\alpha+\beta=3$.
(b)M1	Correct algebra and substitution of their values
A1cso	Correct answer from correct working. Must have seen sum = -3 here if not shown in (a)
(c)	
M1	Factorise to 2 quadratic brackets or 2 linear and one quadratic bracket
A1	Obtain 2 linear and 1 quadratic brackets with 2 of the 3 brackets correct
A1	Third correct bracket Accept $(\alpha^2+\beta^2)$ or $((\alpha+\beta)^2-2\alpha\beta)$
(d)M1	Substitute their values for each of the 3 brackets obtained in (c)
A1	Correct answer as shown or equivalent **exact** value
(e)	
M1	Correct expression for $2\beta^4$ or β^4
A1ft	Substitute their numbers to obtain a numerical expression for β^4 The expression must be exact but need not be simplified **NB** A correct numerical expression for their values implies M1
A1	Correct answer in the required form. p and q need not be shown explicitly.

Type MS · Sum and product of roots

Answer 8 of 21 · Question 10 part a.ii · 5 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Sum and product of roots

Answer 9 of 21 · Question 8 part b · 4 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

8(a)	$x^3+3x^2y+3xy^2+y^3$	B1 [1]
(b)	$\alpha+\beta=-\frac{3}{2}$ **and** $\alpha\beta=\frac{4}{2}$ oe	B1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$	M1
	$\left("-\frac{3}{2}"\right)^3-3("2")\left("-\frac{3}{2}"\right)$ oe	dM1
	$\frac{45}{8}$ oe	A1 [4]
(c)	$\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=$ $\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}=\frac{\left("\frac{45}{8}"\right)}{"2"^2}$	M1
	$\frac{45}{32}$ oe	A1ft
	$\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ $\frac{1}{2}$ oe	B1ft
	$x^2-"\frac{45}{32}"x+"\frac{1}{2}"(=0)$	M1
	$32x^2-45x+16=0$ oe	A1 [5]
		Total 10 marks

Type MS · Sum and product of roots

Answer 9 of 21 · Question 8 part b · 4 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For the correct expansion simplified or un-simplified. For example, allow $x^3+2x^2y+x^2y+xy^2+2xy^2+y^3$
(b)	B1	Product and sum both correct, written explicitly or used later in working.
	M1	An attempt to rearrange their expression from part (a) to achieve as a minimum $\alpha^3+\beta^3=(\alpha+\beta)^3\pm3\alpha\beta(\alpha+\beta)$ **Note:** Accept alternative algebraic arrangements of $\alpha^3+\beta^3$ but please check carefully that the algebra is correct.
	dM1	Substitution of **their** sum and product into an expression which must be of the form $(\alpha+\beta)^3\pm3\alpha\beta(\alpha+\beta)$ **Note:** If they substitute correctly into alternative arrangements check that they are correct.
	A1	For $\frac{45}{8}$ oe
(c)	M1	Correct algebra to obtain $\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}$ **and** substitution of **their** values for the sum and product.
	A1ft	For $\frac{45}{32}$ oe **Note:** only follow through their values for product and sum into **correct algebra.**
	B1ft	For the correct algebra and substitution of their $\alpha\beta=2$ to obtain $\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ ft their $\alpha\beta$
	M1	Use of their sum and product to correctly form an expression as shown, $x^2-"\frac{45}{32}"x+"\frac{1}{2}"(=0)$ Allow missing = 0 for this mark or even = y for this mark.
	A1	Correct equation or any multiple of it, for example , $64x^2-90x+32=0$ The coefficients, must be integers.
	ALT	Starts with $\left(x-\frac{\alpha}{\beta^2}\right)\left(x-\frac{\beta}{\alpha^2}\right)=0\Rightarrow x^2-x\left(\frac{\beta}{\alpha^2}+\frac{\alpha}{\beta^2}\right)+\frac{\beta}{\alpha^2}\times\frac{\alpha}{\beta^2}=0$
	M1	Correct algebra to obtain $\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}$ **and** substitution of **their** values for the sum and product.
	A1ft	For $\frac{45}{32}$ oe **Note:** only follow through their values for product and sum into **correct algebra.**
	B1ft	For the correct algebra and substitution of their $\alpha\beta=2$ to obtain $\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ ft their $\alpha\beta$
	M1	For $x^2-x\left('\frac{45}{32}'\right)+'\frac{1}{2}'=0$ Accept this expression on it's own or even = y for this mark
	A1	Correct equation or any multiple of it, for example , $64x^2-90x+32=0$ The coefficients, must be integers.

Type MS · Sum and product of roots

Answer 9 of 21 · Question 8 part b · 4 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 10 of 21 · Question 7 part a · 1 mark

2025 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
7(a)	$(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3-\alpha^2\beta+\alpha\beta^2+\beta\alpha^2-\alpha\beta^2+\beta^3=\alpha^3+\beta^3$	B1 [1]
(b)	$\alpha+\beta=-4 \quad \alpha\beta=\frac{-k}{2}$	B1
	$\left[\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta\right]$	
	$\alpha^3+\beta^3=(\alpha+\beta)\left((\alpha+\beta)^2-2\alpha\beta-\alpha\beta\right)=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$	M1
	$-94=(-4)\left((-4)^2-3\times\frac{-k}{2}\right)\Rightarrow\frac{47}{2}=16+\frac{3k}{2}\Rightarrow k=5$ *	M1A1 cso [4]
	ALT	
	$\alpha+\beta=-4 \quad \alpha\beta=-\frac{k}{2}$	[B1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$	M1
	$-94=(-4)^3-3\times-\frac{k}{2}\times-4\Rightarrow-30=-6k$	M1
	$k=5$ *	A1]
(c) (i)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=(\alpha+\beta)^2-4\alpha\beta$	M1
	$\alpha-\beta=\sqrt{(-4)^2-4\times-\frac{5}{2}}=\sqrt{26}$ *	M1A1 cso [3]
(ii)	$\alpha^3-\beta^3=(\alpha-\beta)^3+3\alpha\beta(\alpha-\beta)$	M1
	$=\left('\sqrt{26}'\right)^3+3\left(\frac{-5}{2}\right)\left('\sqrt{26}'\right)$	
	$=\frac{37\sqrt{26}}{2}$	A1 [2]
		Total 10 marks

Type MS · Sum and product of roots

Answer 10 of 21 · Question 7 part a · 1 mark

2025 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
7 (a)	B1	A full expansion with no errors.
(b)	B1	For the correct values of the product and sum
	M1	For the correct algebra $\alpha^3+\beta^3$ in a form that values for $\alpha+\beta$ and $\alpha\beta$ [in terms of a] can be substituted.
	M1	For substituting their values of $\alpha+\beta$ and $\alpha\beta$ into their expression for $\alpha^3+\beta^3$ and solving a **linear** equation to find the value of k.
	A1 cso	For the correct value of $k = 5$
	ALT	
	B1	For the correct values of the product and sum
	M1	For the correct algebra $\alpha^3+\beta^3$ in a form that values for $\alpha+\beta$ and $\alpha\beta$ [in terms of a] can be substituted using $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
	M1	For substituting their values of $\alpha+\beta$ and $\alpha\beta$ into their expression for $\alpha^3+\beta^3$ and solving a **linear** equation to find the value of k.
	A1 cso	For the correct value of $k = 5$
(c) (i)	M1	For the correct algebra for $(\alpha-\beta)^2$
	M1	For substituting the correct values for the sum and product and take the positive square root.
	A1 cso	For the correct value with no errors seen.
(ii)	M1	For the correct algebra to find $\alpha^3-\beta^3$ in a form such that the sum and product can be substituted in directly.
	A1	For the correct value of $\alpha^3-\beta^3$

Type MS · Sum and product of roots

Answer 10 of 21 · Question 7 part a · 1 mark

2025 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum and product of roots

Answer 11 of 21 · Question 7 part c.i · 3 marks

2025 NOV · Paper 01 · extract 1 of 3 · find part c.i in this mark scheme extract

Question	Scheme	Marks
7(a)	$(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3-\alpha^2\beta+\alpha\beta^2+\beta\alpha^2-\alpha\beta^2+\beta^3=\alpha^3+\beta^3$	B1 [1]
(b)	$\alpha+\beta=-4 \quad \alpha\beta=\frac{-k}{2}$	B1
	$\left[\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta\right]$	
	$\alpha^3+\beta^3=(\alpha+\beta)\left((\alpha+\beta)^2-2\alpha\beta-\alpha\beta\right)=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$	M1
	$-94=(-4)\left((-4)^2-3\times\frac{-k}{2}\right)\Rightarrow\frac{47}{2}=16+\frac{3k}{2}\Rightarrow k=5$ *	M1A1 cso [4]
	ALT	
	$\alpha+\beta=-4 \quad \alpha\beta=-\frac{k}{2}$	[B1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$	M1
	$-94=(-4)^3-3\times-\frac{k}{2}\times-4\Rightarrow-30=-6k$	M1
	$k=5$ *	A1]
(c) (i)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=(\alpha+\beta)^2-4\alpha\beta$	M1
	$\alpha-\beta=\sqrt{(-4)^2-4\times-\frac{5}{2}}=\sqrt{26}$ *	M1A1 cso [3]
(ii)	$\alpha^3-\beta^3=(\alpha-\beta)^3+3\alpha\beta(\alpha-\beta)$	M1
	$=\left('\sqrt{26}'\right)^3+3\left(\frac{-5}{2}\right)\left('\sqrt{26}'\right)$	
	$=\frac{37\sqrt{26}}{2}$	A1 [2]
		Total 10 marks

Type MS · Sum and product of roots

Answer 11 of 21 · Question 7 part c.i · 3 marks

2025 NOV · Paper 01 · extract 2 of 3 · find part c.i in this mark scheme extract

Part	Mark	Notes
7 (a)	B1	A full expansion with no errors.
(b)	B1	For the correct values of the product and sum
	M1	For the correct algebra $\alpha^3+\beta^3$ in a form that values for $\alpha+\beta$ and $\alpha\beta$ [in terms of a] can be substituted.
	M1	For substituting their values of $\alpha+\beta$ and $\alpha\beta$ into their expression for $\alpha^3+\beta^3$ and solving a **linear** equation to find the value of k.
	A1 cso	For the correct value of $k = 5$
	ALT	
	B1	For the correct values of the product and sum
	M1	For the correct algebra $\alpha^3+\beta^3$ in a form that values for $\alpha+\beta$ and $\alpha\beta$ [in terms of a] can be substituted using $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
	M1	For substituting their values of $\alpha+\beta$ and $\alpha\beta$ into their expression for $\alpha^3+\beta^3$ and solving a **linear** equation to find the value of k.
	A1 cso	For the correct value of $k = 5$
(c) (i)	M1	For the correct algebra for $(\alpha-\beta)^2$
	M1	For substituting the correct values for the sum and product and take the positive square root.
	A1 cso	For the correct value with no errors seen.
(ii)	M1	For the correct algebra to find $\alpha^3-\beta^3$ in a form such that the sum and product can be substituted in directly.
	A1	For the correct value of $\alpha^3-\beta^3$

Type MS · Sum and product of roots

Answer 11 of 21 · Question 7 part c.i · 3 marks

2025 NOV · Paper 01 · extract 3 of 3 · find part c.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum and product of roots

Answer 12 of 21 · Question 7 part c.ii · 2 marks

2025 NOV · Paper 01 · extract 1 of 3 · find part c.ii in this mark scheme extract

Question	Scheme	Marks
7(a)	$(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3-\alpha^2\beta+\alpha\beta^2+\beta\alpha^2-\alpha\beta^2+\beta^3=\alpha^3+\beta^3$	B1 [1]
(b)	$\alpha+\beta=-4 \quad \alpha\beta=\frac{-k}{2}$ $\left[\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta\right]$ $\alpha^3+\beta^3=(\alpha+\beta)\left((\alpha+\beta)^2-2\alpha\beta-\alpha\beta\right)=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$ $-94=(-4)\left((-4)^2-3\times\frac{-k}{2}\right)\Rightarrow\frac{47}{2}=16+\frac{3k}{2}\Rightarrow k=5$ * **ALT** $\alpha+\beta=-4 \quad \alpha\beta=-\frac{k}{2}$ $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ $-94=(-4)^3-3\times-\frac{k}{2}\times-4\Rightarrow-30=-6k$ $k=5$ *	B1 M1 M1A1 cso [4] [B1 M1 M1 A1]
(c) (i)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=(\alpha+\beta)^2-4\alpha\beta$ $\alpha-\beta=\sqrt{(-4)^2-4\times-\frac{5}{2}}=\sqrt{26}$ *	M1 M1A1 cso [3]
(ii)	$\alpha^3-\beta^3=(\alpha-\beta)^3+3\alpha\beta(\alpha-\beta)$ $=\left('\sqrt{26}'\right)^3+3\left(\frac{-5}{2}\right)\left('\sqrt{26}'\right)$ $=\frac{37\sqrt{26}}{2}$	M1 A1 [2]
		Total 10 marks

Type MS · Sum and product of roots

Answer 12 of 21 · Question 7 part c.ii · 2 marks

2025 NOV · Paper 01 · extract 2 of 3 · find part c.ii in this mark scheme extract

Part	Mark	Notes
7 (a)	B1	A full expansion with no errors.
(b)	B1	For the correct values of the product and sum
	M1	For the correct algebra $\alpha^3+\beta^3$ in a form that values for $\alpha+\beta$ and $\alpha\beta$ [in terms of a] can be substituted.
	M1	For substituting their values of $\alpha+\beta$ and $\alpha\beta$ into their expression for $\alpha^3+\beta^3$ and solving a **linear** equation to find the value of k.
	A1 cso	For the correct value of $k = 5$
	ALT	
	B1	For the correct values of the product and sum
	M1	For the correct algebra $\alpha^3+\beta^3$ in a form that values for $\alpha+\beta$ and $\alpha\beta$ [in terms of a] can be substituted using $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
	M1	For substituting their values of $\alpha+\beta$ and $\alpha\beta$ into their expression for $\alpha^3+\beta^3$ and solving a **linear** equation to find the value of k.
	A1 cso	For the correct value of $k = 5$
(c) (i)	M1	For the correct algebra for $(\alpha-\beta)^2$
	M1	For substituting the correct values for the sum and product and take the positive square root.
	A1 cso	For the correct value with no errors seen.
(ii)	M1	For the correct algebra to find $\alpha^3-\beta^3$ in a form such that the sum and product can be substituted in directly.
	A1	For the correct value of $\alpha^3-\beta^3$

Type MS · Sum and product of roots

Answer 12 of 21 · Question 7 part c.ii · 2 marks

2025 NOV · Paper 01 · extract 3 of 3 · find part c.ii in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum and product of roots

Answer 13 of 21 · Question 5 part a · 2 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

5 (a)	$(\alpha+\beta)^3 = \alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3 \Rightarrow \alpha^3+\beta^3$ $=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$*	M1A1 cso [2]
(b)	$\alpha+\beta=-\frac{3}{2} \qquad \alpha\beta=\frac{6}{2}=3$ $\left[\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right]$ $\alpha^3+\beta^3=\left(-\frac{3}{2}\right)^3-3\times3\times\left(-\frac{3}{2}\right)=\frac{81}{8}$	B1 B1 [2]
(c)	$(\alpha^2+\beta^2)^2=\alpha^4+2\alpha^2\beta^2+\beta^4 \Rightarrow \alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2(\alpha\beta)^2$*	M1A1 cso [2]
(d)	$\alpha^2+\beta^2=\left(-\frac{3}{2}\right)^2-2\times3=-\frac{15}{4}$ Sum $\quad(\alpha^3-\beta)+(\beta^3-\alpha)=\alpha^3+\beta^3-(\alpha+\beta)=\frac{81}{8}-\left(-\frac{3}{2}\right)=\frac{93}{8}$ Product $\quad(\alpha^3-\beta)\times(\beta^3-\alpha)=(\alpha\beta)^3-(\alpha^4+\beta^4)+\alpha\beta$ $=27-\left[\left(-\frac{15}{4}\right)^2-2\times3^2\right]+3=\frac{543}{16}$ Equation $\quad x^2-\frac{93}{8}x+\frac{543}{16}=0\Rightarrow16x^2-186x+543=0$	B1 B1 M1 A1 M1A1 [6]
		Total 12 marks

Type MS · Sum and product of roots

Answer 13 of 21 · Question 5 part a · 2 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

(d)	
B1	$\alpha^2+\beta^2=-\frac{15}{4}$ $\left(\text{May be implied by } \alpha^4+\beta^4=-\frac{63}{16}\right)$
B1	$\left(\alpha^3-\beta\right)+\left(\beta^3-\alpha\right)=\frac{93}{8}$
M1	$\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=\left(\alpha\beta\right)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$
A1	$\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=\frac{543}{16}$
M1	$x^2-\text{'sum'}\,x+\text{'product'}\left(=0\right)$
A1	$16x^2-186x+543=0$

Type MS · Sum and product of roots

Answer 13 of 21 · Question 5 part a · 2 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 14 of 21 · Question 5 part b · 2 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

5 (a)	$(\alpha+\beta)^3 = \alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3 \Rightarrow \alpha^3+\beta^3$ $=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$*	M1A1 cso [2]
(b)	$\alpha+\beta=-\frac{3}{2} \qquad \alpha\beta=\frac{6}{2}=3$ $\left[\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right]$ $\alpha^3+\beta^3=\left(-\frac{3}{2}\right)^3-3\times3\times\left(-\frac{3}{2}\right)=\frac{81}{8}$	B1 B1 [2]
(c)	$\left(\alpha^2+\beta^2\right)^2=\alpha^4+2\alpha^2\beta^2+\beta^4 \Rightarrow \alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2(\alpha\beta)^2$*	M1A1 cso [2]
(d)	$\alpha^2+\beta^2=\left(-\frac{3}{2}\right)^2-2\times3=-\frac{15}{4}$ Sum $\left(\alpha^3-\beta\right)+\left(\beta^3-\alpha\right)=\alpha^3+\beta^3-(\alpha+\beta)=\frac{81}{8}-\left(-\frac{3}{2}\right)=\frac{93}{8}$ Product $\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=(\alpha\beta)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$ $=27-\left[\left(-\frac{15}{4}\right)^2-2\times3^2\right]+3=\frac{543}{16}$ Equation $x^2-\frac{93}{8}x+\frac{543}{16}=0 \Rightarrow 16x^2-186x+543=0$	B1 B1 M1 A1 M1A1 [6]
		Total 12 marks

Type MS · Sum and product of roots

Answer 14 of 21 · Question 5 part b · 2 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(d)	
B1	$\alpha^2+\beta^2=-\frac{15}{4}$ $\left(\text{May be implied by } \alpha^4+\beta^4=-\frac{63}{16}\right)$
B1	$\left(\alpha^3-\beta\right)+\left(\beta^3-\alpha\right)=\frac{93}{8}$
M1	$\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=\left(\alpha\beta\right)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$
A1	$\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=\frac{543}{16}$
M1	$x^2-\text{'sum'}\,x+\text{'product'}\left(=0\right)$
A1	$16x^2-186x+543=0$

Type MS · Sum and product of roots

Answer 14 of 21 · Question 5 part b · 2 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 15 of 21 · Question 5 part c · 2 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

5 (a)	$(\alpha+\beta)^3 = \alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3 \Rightarrow \alpha^3+\beta^3$ $=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$*	M1A1 cso [2]
(b)	$\alpha+\beta=-\frac{3}{2} \qquad \alpha\beta=\frac{6}{2}=3$ $\left[\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right]$ $\alpha^3+\beta^3=\left(-\frac{3}{2}\right)^3-3\times3\times\left(-\frac{3}{2}\right)=\frac{81}{8}$	B1 B1 [2]
(c)	$\left(\alpha^2+\beta^2\right)^2=\alpha^4+2\alpha^2\beta^2+\beta^4 \Rightarrow \alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2(\alpha\beta)^2$*	M1A1 cso [2]
(d)	$\alpha^2+\beta^2=\left(-\frac{3}{2}\right)^2-2\times3=-\frac{15}{4}$ Sum $\left(\alpha^3-\beta\right)+\left(\beta^3-\alpha\right)=\alpha^3+\beta^3-(\alpha+\beta)=\frac{81}{8}-\left(-\frac{3}{2}\right)=\frac{93}{8}$ Product $\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=(\alpha\beta)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$ $=27-\left[\left(-\frac{15}{4}\right)^2-2\times3^2\right]+3=\frac{543}{16}$ Equation $x^2-\frac{93}{8}x+\frac{543}{16}=0 \Rightarrow 16x^2-186x+543=0$	B1 B1 M1 A1 M1A1 [6]
		Total 12 marks

Type MS · Sum and product of roots

Answer 15 of 21 · Question 5 part c · 2 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

(d)	
B1	$\alpha^2+\beta^2=-\frac{15}{4}$ $\left(\text{May be implied by } \alpha^4+\beta^4=-\frac{63}{16}\right)$
B1	$\left(\alpha^3-\beta\right)+\left(\beta^3-\alpha\right)=\frac{93}{8}$
M1	$\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=\left(\alpha\beta\right)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$
A1	$\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=\frac{543}{16}$
M1	$x^2-\text{'sum'}x+\text{'product'}(=0)$
A1	$16x^2-186x+543=0$

Type MS · Sum and product of roots

Answer 15 of 21 · Question 5 part c · 2 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 16 of 21 · Question 10 part d · 1 mark

2024 NOV · Paper 01 · find part d in this mark scheme extract

10 (a)	$\left(3-9\left(\frac{4}{9}x+x^2\right)\right.$ or $\left.-9\left(-\frac{1}{3}+\frac{4}{9}x+x^2\right)\right)$ $3-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}\right]$ or $-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}-\frac{1}{3}\right]$ or $-1\left[9\left(x+\frac{2}{9}\right)^2-\frac{36}{81}-3\right]$ or $3+(1)\left[-9\left(x+\frac{2}{9}\right)^2+\frac{36}{81}\right]$ $\frac{31}{9}-9\left(x+\frac{2}{9}\right)^2$ $A=\frac{31}{9}$ $B=9$ $C=\frac{2}{9}$ oe	M1 A1 A1 A1 [4]
(b)	$"\frac{31}{9}"$	B1ft [1]
(c)	$\alpha+\beta=-\frac{4}{9}$ $\alpha\beta=-\frac{3}{9}$ oe $\frac{3\alpha}{\beta}+\frac{3\beta}{\alpha}\left(=\frac{3\left(\alpha^2+\beta^2\right)}{\alpha\beta}\right)=\frac{3\left((\alpha+\beta)^2-2\alpha\beta\right)}{\alpha\beta}$ $\frac{3\left[\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)\right]}{"-\frac{3}{9}"}$ $\left(=-\frac{70}{9}\right)$ $\left(\frac{3\alpha}{\beta}\times\frac{3\beta}{\alpha}=\right)9$ $x^2+"\frac{70}{9}"x+"9"\,(=0)$ $9x^2+70x+81=0$	B1 M1 A1ft B1 M1 A1cso [6]
(d)	$\left((x+y)^3=\right)x^3+3x^2y+3xy^2+y^3=x^3+y^3+3xy(x+y)$	B1*cso [1]
(e)	$(\alpha^2-\beta+\beta^2-\alpha=)$ $(\alpha+\beta)^2-2\alpha\beta-(\alpha+\beta)$ $\left((\alpha+\beta)^2=(\alpha+\beta)^2-2\alpha\beta=\frac{70}{81}\right)$ $\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)$ $\left(=\frac{106}{81}\right)$ **candidates may also transfer a value for $\alpha^2+\beta^2$ from working in part (c)** $\left[(\alpha^2-\beta)(\beta^2-\alpha)=\alpha^2\beta^2-\beta^3-\alpha^3+\alpha\beta\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha^3+\beta^3)$ or $(\alpha\beta)^2+\alpha\beta-\alpha^3-\beta^3$ $\left[(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha+\beta)^3+3\alpha\beta(\alpha+\beta)$ or $(\alpha\beta)^2+\alpha\beta-\left((\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right)$ $\left[=\left("-\frac{3}{9}"\right)^2+\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)^3+3\left("-\frac{3}{9}"\right)\left("-\frac{4}{9}"\right)\left(=\frac{226}{729}\right)\right]$ $"\frac{106}{81}"=-\frac{q}{3}$ or $"\frac{226}{729}"=\frac{r}{3}$ $q=-\frac{106}{27}$ and $r=\frac{226}{243}$	M1 A1ft M1 M1 dddM1 A1 [6]
		Total 18 marks

Type MS · Sum and product of roots

Answer 17 of 21 · Question 5 part a.i · 2 marks

2022 JUN · Paper 02 · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Sum and product of roots

Answer 18 of 21 · Question 8 part b.i · 0 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b.i in this mark scheme extract

8 (a)	$\left(\frac{dy}{dx}=\right)4x-4$	M1
	Gradient of $l=8$	B1
	Gradient of normal $=-\frac{1}{8}$	B1
	$4x-4=-\frac{1}{8}\Rightarrow x=\frac{31}{32}$	M1A1 [5]
(b) (i)	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)$	B1*cso [1]
(ii)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2\alpha^2\beta^2=\left((\alpha+\beta)^2-2\alpha\beta\right)^2-2(\alpha\beta)^2$	B1*cso [1]
(c)	$\alpha+\beta=2\quad\alpha\beta=\frac{9}{2}$ oe	B1
	$\left(\alpha^3-\beta+\beta^3-\alpha=\alpha^3+\beta^3-(\alpha+\beta)\right)$	
	$=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)-(\alpha+\beta)$	M1
	(sum of roots $=$)$2^3-3\left(\frac{9}{2}\right)(2)-2(=-21)$	A1
	$\left[\left(\alpha^3-\beta\right)\left(\beta^3-\alpha\right)=\right]\alpha^3\beta^3-\alpha^4-\beta^4+\alpha\beta$	M1
	$=(\alpha\beta)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$	
	Now, $\left(\alpha^4+\beta^4\right)=\left((2)^2-2\times\frac{9}{2}\right)^2-2\times\left(\frac{9}{2}\right)^2=\left(-\frac{31}{2}\right)$	A1ft
	(product of roots $=$) $=\left(\frac{9}{2}\right)^3-\left(-\frac{31}{2}\right)+\frac{9}{2}\left(=\frac{889}{8}\right)$	A1
	$x^2-"-21"x+"\frac{889}{8}"$	M1
	$8x^2+168x+889=0$	A1 [8]
		Total 15 marks

Type MS · Sum and product of roots

Answer 18 of 21 · Question 8 part b.i · 0 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b.i in this mark scheme extract

Part	Mark	Additional Guidance
8 (a)	M1	For a correct differentiation.
	B1	For 8, identified as the gradient or used to calculate gradient of normal
	B1	For $-\frac{1}{8}$ stated as gradient of normal or used in subsequent work
	M1	For correctly solving their $ax+b=-\frac{1}{8}$
	A1	For $\frac{31}{32}$
(b) (i)	B1*cso	Complete and full algebra to show the given identity
(ii)	B1*cso	Complete and full algebra to show the given identity
(c)	B1	Correct values for $\alpha+\beta$ **and** $\alpha\beta$. Award if values not seen explicitly but embedded in the sum/product calculations for the new equation.
	M1	**Sum:** Using the result from part (b) and correctly manipulating to prepare for substitution of $\alpha+\beta$ **and** $\alpha\beta$
	A1	For -21
	M1	**Product:** For the correct algebraic expansion shown.
	A1ft	Correct substitution of their values into $\alpha^4+\beta^4$ within the product of roots
	A1	For $\frac{889}{8}$
	M1	Use $x^2-(\text{their sum of their roots})x+\text{their product of roots}$ "= 0" may be missing.
	A1	Correct equation oe with integer coefficients.

Type MS · Sum and product of roots

Answer 18 of 21 · Question 8 part b.i · 0 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 19 of 21 · Question 8 part b.ii · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b.ii in this mark scheme extract

8 (a)	$\left(\frac{dy}{dx}=\right)4x-4$	M1
	Gradient of $l=8$	B1
	Gradient of normal $=-\frac{1}{8}$	B1
	$4x-4=-\frac{1}{8}\Rightarrow x=\frac{31}{32}$	M1A1 [5]
(b) (i)	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)$	B1*cso [1]
(ii)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2\alpha^2\beta^2=\left((\alpha+\beta)^2-2\alpha\beta\right)^2-2(\alpha\beta)^2$	B1*cso [1]
(c)	$\alpha+\beta=2\quad\alpha\beta=\frac{9}{2}$ oe	B1
	$\left(\alpha^3-\beta+\beta^3-\alpha=\alpha^3+\beta^3-(\alpha+\beta)\right)$	
	$=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)-(\alpha+\beta)$	M1
	(sum of roots $=$)$2^3-3\left(\frac{9}{2}\right)(2)-2(=-21)$	A1
	$\left[\left(\alpha^3-\beta\right)\left(\beta^3-\alpha\right)=\right]\alpha^3\beta^3-\alpha^4-\beta^4+\alpha\beta$	M1
	$=(\alpha\beta)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$	
	Now, $\left(\alpha^4+\beta^4\right)=\left((2)^2-2\times\frac{9}{2}\right)^2-2\times\left(\frac{9}{2}\right)^2=\left(-\frac{31}{2}\right)$	A1ft
	(product of roots $=$) $=\left(\frac{9}{2}\right)^3-\left(-\frac{31}{2}\right)+\frac{9}{2}\left(=\frac{889}{8}\right)$	A1
	$x^2-"-21"x+"\frac{889}{8}"$	M1
	$8x^2+168x+889=0$	A1 [8]
		Total 15 marks

Type MS · Sum and product of roots

Answer 19 of 21 · Question 8 part b.ii · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b.ii in this mark scheme extract

Part	Mark	Additional Guidance
8 (a)	M1	For a correct differentiation.
	B1	For 8, identified as the gradient or used to calculate gradient of normal
	B1	For $-\frac{1}{8}$ stated as gradient of normal or used in subsequent work
	M1	For correctly solving their $ax+b=-\frac{1}{8}$
	A1	For $\frac{31}{32}$
(b) (i)	B1*cso	Complete and full algebra to show the given identity
(ii)	B1*cso	Complete and full algebra to show the given identity
(c)	B1	Correct values for $\alpha+\beta$ **and** $\alpha\beta$. Award if values not seen explicitly but embedded in the sum/product calculations for the new equation.
	M1	**Sum:** Using the result from part (b) and correctly manipulating to prepare for substitution of $\alpha+\beta$ **and** $\alpha\beta$
	A1	For -21
	M1	**Product:** For the correct algebraic expansion shown.
	A1ft	Correct substitution of their values into $\alpha^4+\beta^4$ within the product of roots
	A1	For $\frac{889}{8}$
	M1	Use $x^2-(\text{their sum of their roots})x+\text{their product of roots}$ "= 0" may be missing.
	A1	Correct equation oe with integer coefficients.

Type MS · Sum and product of roots

Answer 19 of 21 · Question 8 part b.ii · 2 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum and product of roots

Answer 20 of 21 · Question 11 part a.i · 0 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

11 (a)	(i) $2\alpha\beta = 3 \Rightarrow q = \alpha\beta \Rightarrow q = \frac{3}{2}$ (ii) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$, $\left[\alpha^2+\beta^2 = \frac{k^2-6k-3}{4}\right]$ $(\alpha+\beta)^2 = \frac{k^2-6k-3}{4} + 3 = \frac{k^2-6k+9}{4} = \frac{(k-3)^2}{4}$ $\Rightarrow \alpha+\beta = \frac{k-3}{2} \Rightarrow p = \frac{k-3}{2}$	B1 M1A1 M1A1 [5]
(b)	$7\left(\frac{3}{2}\right) = 3\left(\frac{k-3}{2}\right) \Rightarrow 7 = k-3 \Rightarrow k = 10$	M1A1 [2]
(c)	$\alpha+\beta = \frac{10-3}{2} = \frac{7}{2}$ Sum: $\frac{\alpha}{\alpha+\beta} + \frac{\beta}{\alpha+\beta} = \frac{\alpha+\beta}{\alpha+\beta} = 1$ Product: $\frac{\alpha}{\alpha+\beta} \times \frac{\beta}{\alpha+\beta} = \frac{\alpha\beta}{(\alpha+\beta)^2} = \frac{\frac{3}{2}}{\left(\frac{7}{2}\right)^2} = \frac{6}{49}$ Equation: $x^2 - x + \frac{6}{49}$ $(=0)$ $49x^2 - 49x + 6 = 0$	M1 B1 M1 M1 A1 [5]

(a) (i)	B1	$q = \frac{3}{2}$
(ii)	M1	State or use any correct arrangement of $(\alpha+\beta)^2 = \alpha^2+\beta^2+2\alpha\beta$ and substitute $\alpha\beta = \frac{3}{2}$ and $\alpha^2+\beta^2 = \frac{k^2-6k-3}{4}$ oe p may be seen instead of $\alpha+\beta$ at any stage of the working. Condone $-p = \alpha+\beta$ for this mark only.
	A1	Any correct expression for $(\alpha+\beta)^2$ or $4(\alpha+\beta)^2$ in terms of k
	M1	Correct method to find an expression for p in terms of k from their expression for $(\alpha+\beta)^2$ or $4(\alpha+\beta)^2$. Ignore negative square root. Must identify $p = \alpha+\beta$.
	A1	A correct expression for p in terms of k. The question does not demand simplification, but if it is left as a square root there must be an indication that p must be the positive root (ie no $\pm$ present).

Type MS · Sum and product of roots

Answer 20 of 21 · Question 11 part a.i · 0 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

(b)	M1	Substitute $\alpha\beta = \frac{3}{2}$ and their expression for $\alpha+\beta$ from (a)(ii) into $7\alpha\beta = 3(\alpha+\beta)$ and obtain an equation in k
	A1	$k = 10$ Do not accept if there is an additional root, such as $k = -4$.
(c)	M1	Find a value for $\alpha+\beta$. Need not be simplified. Follow through their k and their expression for p.
	B1	Show or state sum of roots = 1 (stated or explicitly used).
	M1	Write product of roots as $\frac{3/2}{(7/2)^2}$, ft their $\alpha+\beta = \frac{7}{2}$.
	M1	Use equation is $x^2 -$ (sum of roots) + (product of roots) $(=0)$ $(=0$ may be missing). There must be some attempt to substitute values, not just expressions in terms of α and β. May be implied by an equation with fractions correctly cleared.
	A1	$49x^2 - 49x + 6 = 0$ Correct equation with integer coefficients. Must have = 0.

Type MS · Sum and product of roots

Answer 21 of 21 · Question 8 part a · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

8(a)	$x^3+3x^2y+3xy^2+y^3$	B1 [1]
(b)	$\alpha+\beta=-\frac{3}{2}$ **and** $\alpha\beta=\frac{4}{2}$ oe	B1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$	M1
	$\left("-\frac{3}{2}"\right)^3-3("2")\left("-\frac{3}{2}"\right)$ oe	dM1
	$\frac{45}{8}$ oe	A1 [4]
(c)	$\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=$ $\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}=\frac{\left("\frac{45}{8}"\right)}{"2"^2}$	M1
	$\frac{45}{32}$ oe	A1ft
	$\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ $\frac{1}{2}$ oe	B1ft
	$x^2-"\frac{45}{32}"x+"\frac{1}{2}"(=0)$	M1
	$32x^2-45x+16=0$ oe	A1 [5]
		Total 10 marks

Type MS · Sum and product of roots

Answer 21 of 21 · Question 8 part a · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For the correct expansion simplified or un-simplified. For example, allow $x^3+2x^2y+x^2y+xy^2+2xy^2+y^3$
(b)	B1	Product and sum both correct, written explicitly or used later in working.
	M1	An attempt to rearrange their expression from part (a) to achieve as a minimum $\alpha^3+\beta^3=(\alpha+\beta)^3\pm3\alpha\beta(\alpha+\beta)$ **Note:** Accept alternative algebraic arrangements of $\alpha^3+\beta^3$ but please check carefully that the algebra is correct.
	dM1	Substitution of **their** sum and product into an expression which must be of the form $(\alpha+\beta)^3\pm3\alpha\beta(\alpha+\beta)$ **Note:** If they substitute correctly into alternative arrangements check that they are correct.
	A1	For $\frac{45}{8}$ oe
(c)	M1	Correct algebra to obtain $\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}$ **and** substitution of **their** values for the sum and product.
	A1ft	For $\frac{45}{32}$ oe **Note:** only follow through their values for product and sum into **correct algebra.**
	B1ft	For the correct algebra and substitution of their $\alpha\beta=2$ to obtain $\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ ft their $\alpha\beta$
	M1	Use of their sum and product to correctly form an expression as shown, $x^2-"\frac{45}{32}"x+"\frac{1}{2}"(=0)$ Allow missing = 0 for this mark or even = y for this mark.
	A1	Correct equation or any multiple of it, for example , $64x^2-90x+32=0$ The coefficients, must be integers.
	ALT Starts with $\left(x-\frac{\alpha}{\beta^2}\right)\left(x-\frac{\beta}{\alpha^2}\right)=0\Rightarrow x^2-x\left(\frac{\beta}{\alpha^2}+\frac{\alpha}{\beta^2}\right)+\frac{\beta}{\alpha^2}\times\frac{\alpha}{\beta^2}=0$	
	M1	Correct algebra to obtain $\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}$ **and** substitution of **their** values for the sum and product.
	A1ft	For $\frac{45}{32}$ oe **Note:** only follow through their values for product and sum into **correct algebra.**
	B1ft	For the correct algebra and substitution of their $\alpha\beta=2$ to obtain $\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ ft their $\alpha\beta$
	M1	For $x^2-x\left('\frac{45}{32}'\right)+'\frac{1}{2}'=0$ Accept this expression on it's own or even = y for this mark
	A1	Correct equation or any multiple of it, for example , $64x^2-90x+32=0$ The coefficients, must be integers.

Type MS · Sum and product of roots

Answer 21 of 21 · Question 8 part a · 1 mark

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks